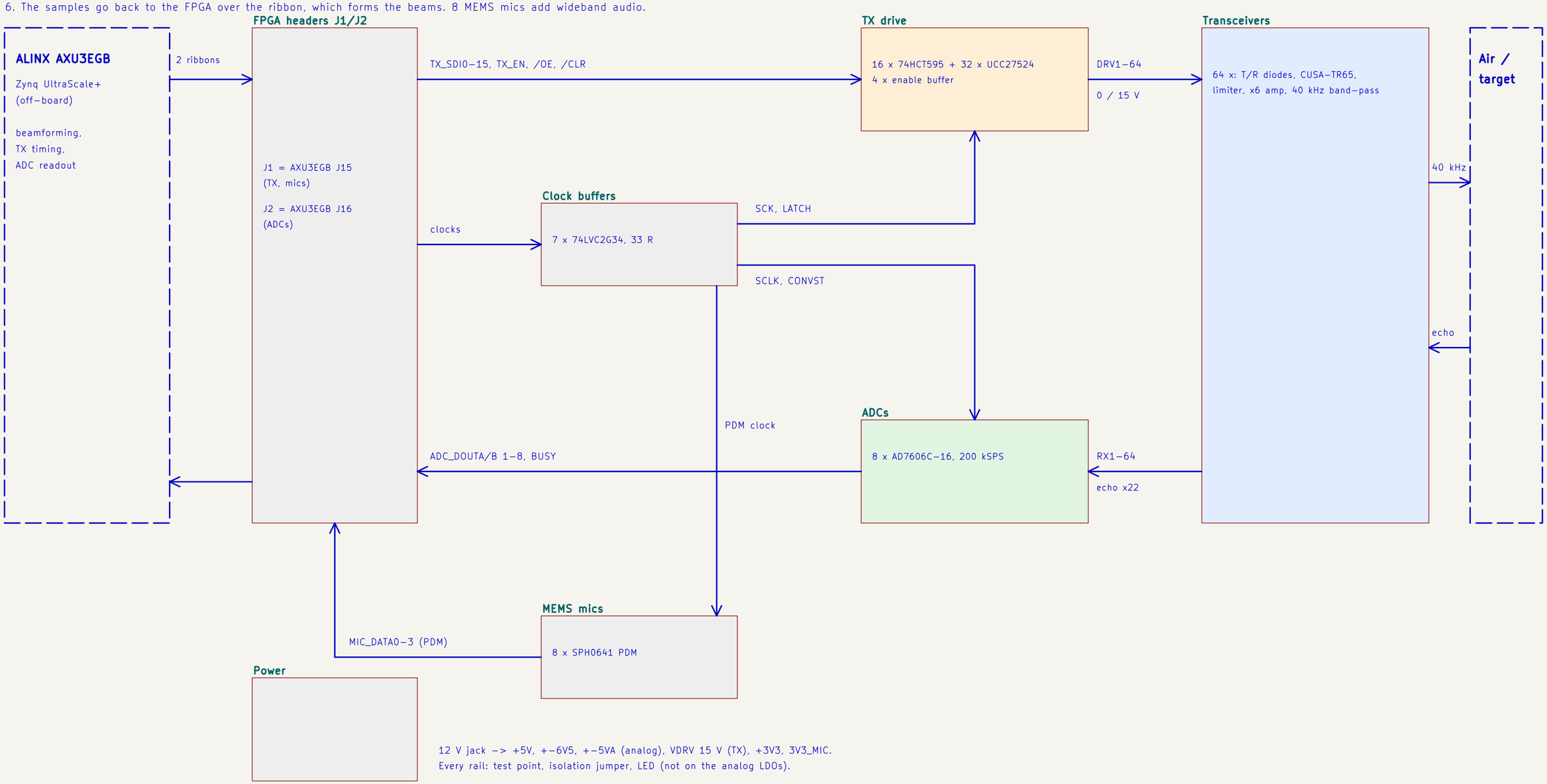
Hybrid 8x8 ultrasonic AESA: how the signal flows

1. The FPGA streams one TX bit per element (16 serial lines, 10 MHz) and the clocks.
2. TX drive: shift registers latch the bits; gate drivers switch each element between 0 and 15 V.
3. Transceivers: the 64 CUSA-TR65 elements emit 40 kHz; echoes return to the same elements.
4. The diode T/R switch and limiter protect the receiver; amplifier + 40 kHz band-pass give x22.
5. ADCs: 8 x AD7606C-16 sample all 64 channels at 200 kSPS (raw waveform, amplitude and phase).
6. The samples go back to the FPGA over the ribbon, which forms the beams. 8 MEMS mics add wideband audio.



Click a block to open it. Repeated parts are one sheet each: channel x64, adc x8, tx_group x16, en_buf x4, clk_buf x7, mic x8.

Hybrid 8x8 ultrasonic AESA		
Sheet: /		
File: aesa.kicad_sch		
Title: Hybrid 8x8 ultrasonic AESA: block diagram		
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